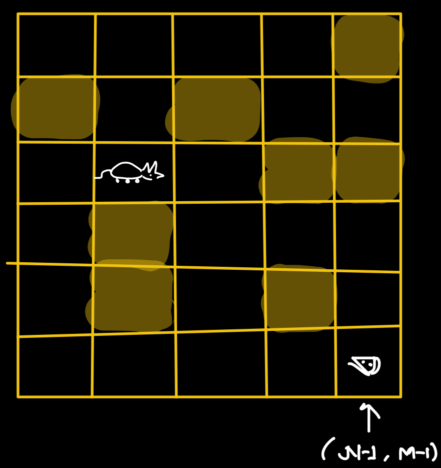
Performance = Potential − interferences
(distractions)

Facebook
Amazon
MS
Flipkart
Samsung
Goldman Sachs
⋮
Cleartrip

Q Rat in a maze

Given
- a maze (2D matrix $N \times M$)
- position of rat $(x, y)$

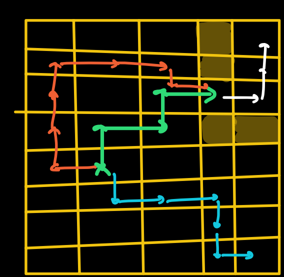
Return true if there exists a path from rat to cheese.




mat[i][j] ⟶ 1 blocked
mat[i][j] ⟶ 0 open

$(x, y)$ ⟶ $(x-1, y)$, $(x, y-1)$, $(x, y+1)$, $(x+1, y)$

$(x-1, y)$ ⟶ $(x-2, y)$, $(x-1, y-1)$, $(x-1, y+1)$, $(x, y)$

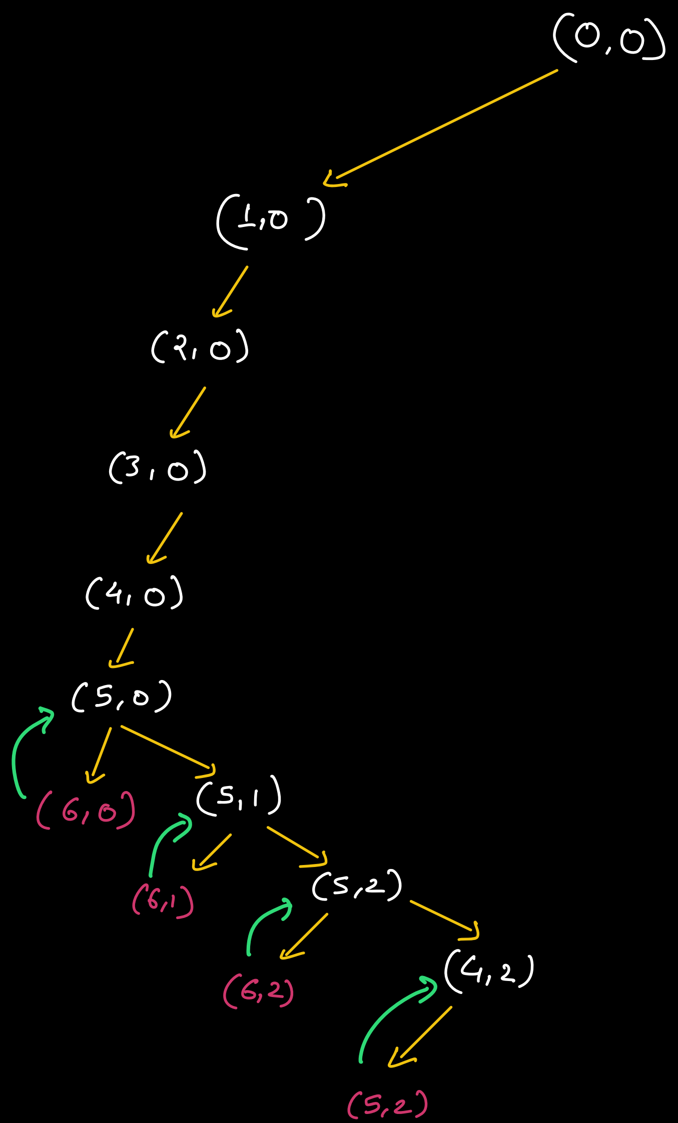


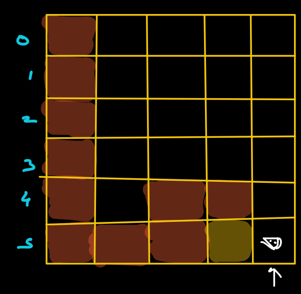
```
boolean   mazeSolver ( mat, N, M, x, y ) {

    if ( x<0 || x>=N || y<0 || y>=M ) {
            ret false;
    if ( mat[x][y] == 1   ||  mat[x][y] == 2 )
            ret false;
    if ( x = N-1   &&  y == M-1 )
            ret true;
    mat[x][y] = 2;

    ret   mazeSolver ( mat, N, M, x+1, y )
                     ||
          mazeSolver ( mat, N, M, x, y+1 )
                     ||
          mazeSolver ( mat, N, M, x-1, y )
                     ||
          mazeSolver ( mat, N, M, x, y-1 )
}
```

$f(N) = f(n-1) + f(n-2)$

$2^N$

TC: O(NM)
SC: ?   HW




R

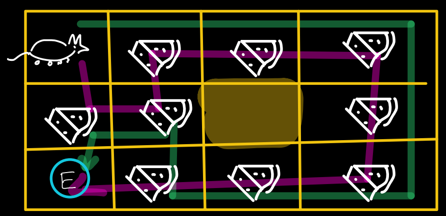
Google
(Hard)

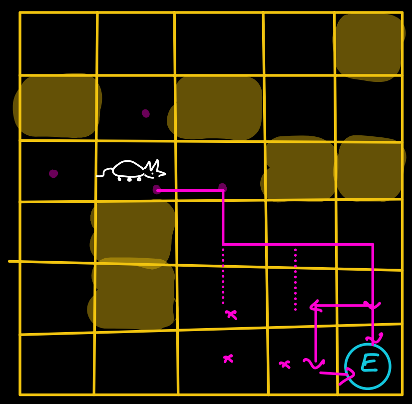
# Rat in a maze

Given

- start point of rat $(s_i, s_j)$
- end point of rat $(e_i, e_j)$
- cells with cheeze 0
- blocked cells 1
- empty cells 2

Count the no of paths from $(s_i, s_j)$ to $(e_i, e_j)$ such that the rat can eat all the cheeze available in the maze without stepping on same cell more than once (in a single path



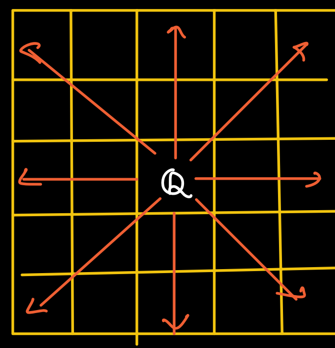



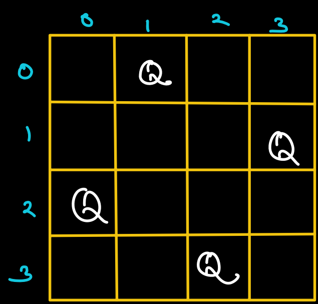
```
int   count Path (mat, N, M, s_i, s_j , e_i , e_j , totalCheeze, curCheeze) {

        if (s_i < 0 || s_i >= N || s_j < 0 || s_j >= M) {
                ret 0;
        }
        if (mat[s_i][s_j] == 1 || mat[s_i][s_j] == -1)
                ret 0;
        if ( s_i == e_i && s_j == e_j) {
                if (totalCheeze == curCheeze) {
                        ret 1;
                }

                ret 0;
        }
    curCheeze = mat[s_i][s_j] == 0 ? (curCheeze + 1) : curCheeze ;
    temp = mat[s_i][s_j]
    mat[s_i][s_j] = -1 ;     // mark visited
    int ans = count Path (mat, N, M, s_i+1, s_j, e_i, e_j, totalCheeze, curCheeze)
                        +
              count Path (mat, N, M, s_i, s_j+1, e_i, e_j, totalCheeze, curCheeze)
                        +
              count Path (mat, N, M, s_i-1, s_j, e_i, e_j, totalCheeze, curCheeze)
                        +
              count Path (mat, N, M, s_i, s_j-1, e_i, e_j, totalCheeze, curCheeze)

    mat [s_i][s_j] = temp ;
    // Reset Curr Cheeze        <------------ Check if req ?
    ret ans;
}
```

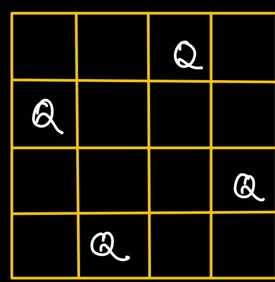
Google
Facebook
Uber
Amazon
MS
Atlassian
Goldman Sachs
Adobe
Twitter
PayTM
Paypal
Ola
Dunzo
Flipkart
Myntra
⋮

# Q N-Queen Problem

Given a N×N chess & N queen
Arrange the queens in the
board such that no queen
can target any other queen.




HW